CUSTOMER AND SALES FORECASTING REPORT

1. Overview and Objectives

This project analyzes customer purchasing behavior and sales performance for a multi-category fashion retailer (**Clothing, Accessories, Footwear, Outerwear**) and forecasts the next six months of sales using historical data and Python-based time-series modeling. The analysis is implemented in Power BI for interactive exploration and supported by a forecasting dataset (**forecast_all_categories**) that provides monthly predicted sales by category.

The core business questions addressed are:

- Who are the most valuable customer segments by age, gender, location, and subscription status?

- Which product categories, sizes, and colors drive revenue today?

- How will category-wise sales evolve over the next six months, and where should inventory and marketing be focused?

2. Current Customer and Revenue Profile

2.1 Global KPIs

The dataset contains **3,900 customers** generating a total purchase amount of about $233K with an average purchase value of $59.76 The **average review rating is 3.75**, indicating moderate satisfaction and clear scope to improve customer experience and product quality.

2.2 Subscription and Gender Distribution

Around 73% of customers are **non-subscribers (2,847)** and only 27% are **subscribers (1,053).** Gender is nearly balanced, with **53.75% female** and **46.25% male** customers.

2.3 Age-Group Contribution

Revenue is well distributed across age groups: Young Adults lead with $62K, followed by Middle-aged ($59K), Adults ($56K), and Seniors ($56K). Order counts follow a similar pattern: Young Adults place 1,028 orders, Middle-aged 986, Seniors 944, and Adults 942.

2.4 Geographic and Payment Behavior

Top locations by purchase value are states such as Montana, Illinois, California, Idaho, and Nevada, each contributing around 5.5K–5.8K in revenue. Payment methods are well

diversified: PayPal, Cash, and Credit Card each account for about 17–18% of transactions, with Venmo, Bank Transfer, and Debit Card each close behind at 15–16%.

3. Product, Category, and Customer Behavior Insights

3.1 Category Performance Today

Clothing is the leading category, generating $104K in revenue from 1,737 orders. Accessories follow with $74K revenue and 1,240 orders. Footwear contributes $36K from 599 orders, and Outerwear $19K from 324 orders.

3.2 Size, Color, and Item Preferences

Size-wise revenue is highest for M ($105K), followed by L ($62K), S ($40K), and XL ($26K). The top colors by purchase are Green, Teal, Yellow, Olive, and Cyan, each with roughly 10K–11K in revenue. The most purchased items include Blouse, Shirt, Dress, Pants, and Jewelry, all clustered around 10K in revenue.

3.3 Gender–Category Dynamics

For each category (Clothing, Accessories, Footwear, Outerwear), male customers contribute roughly 60–70% of sales, and female customers about 30–40%. In absolute terms, **male customers spend more across all categories**, **despite only being 46.25% of the base**.

3.4 Discount, Rating, and Frequency Behavior

A higher share of sales (about 57%) occurs without discounts, while 43% involves discounted purchases. Purchase frequency is spread across patterns like Weekly, Bi-Weekly, Monthly, Quarterly, etc., each around 13–15%, showing a mix of regular and occasional shoppers. The relationship between purchase amount and review rating is moderately positive: higher ratings correlate with slightly higher purchase amounts.

4. Seasonal and Temporal Sales Patterns

4.1 Season-Wise Revenue

Seasonal revenue ranges from roughly 55–60K per season, with Summer and Winter slightly ahead of Fall and Spring. When broken down by category, Clothing and Accessories show strong sales in all seasons, while Footwear and Outerwear have relatively lower but steady contributions.

4.2 Age and Season Interactions

Age-group share of category revenue remains relatively stable across seasons, with Young Adults and Middle-aged segments consistently slightly higher. This suggests that younger and middle-aged customers are consistently engaged, rather than peaking only in specific seasons.

5. Forecasted Category-Wise Sales (Next Six Months)

5.1 Total Predicted Sales by Category (Six-Month Sum)

Clothing is expected to contribute about 41.54% of predicted sales over the next six months, followed by Accessories at 33.28%, Footwear at 15.26%, and Outerwear at 9.92%. This confirms Clothing as the core future category, with Accessories as a strong secondary driver.

5.2 Monthly Max Predicted Sales by Category

For each of the next six months (January–June), the maximum predicted monthly sales by category show stable high demand for Clothing (around 3,600–4,000 each month) and Accessories (around 2,900–3,200). Footwear and Outerwear remain smaller but steady, with monthly maxima around 1,300–1,500 for Footwear and 850–980 for Outerwear.

6. Recommendations and Next Steps

1. Strengthen subscription program by designing targeted subscription campaigns for high-value segments (Young Adults and Middle-aged males in Clothing and Accessories) to convert part of the 73% non-subscriber base.

2. Deepen focus on Clothing and Accessories by allocating the majority of marketing spend and inventory budget to these categories, particularly in medium and large sizes and top colors.

3. Optimize male-oriented product lines since male customers spend more across all categories, building premium collections, bundles, and loyalty perks specifically tailored to them.

4. Improve customer experience to raise ratings above 4.0 by using feedback from reviews, returns, and customer support tickets.

5. Leverage stable seasonality and forecasts for inventory and cash-flow planning, updating the model periodically as new data comes in.